

Exercise 21D — Full Solutions

Questions 1–8 (complete)

Higher derivatives and Maclaurin series

Colour key: *green* = verification check | *blue* = alternative method.

Question 1

Find the fourth derivative of each function.

(a) $y = e^{x^2}$:

$$y' = 2x e^{x^2}, \quad y'' = (2 + 4x^2)e^{x^2}, \quad y''' = (8x^3 + 12x)e^{x^2},$$

$$y^{(4)} = (16x^4 + 48x^2 + 12)e^{x^2} = 4(4x^4 + 12x^2 + 3)e^{x^2}.$$

(b) $y = x \cos x$:

$$y' = \cos x - x \sin x, \quad y'' = -2 \sin x - x \cos x, \quad y''' = -3 \cos x + x \sin x,$$

$$y^{(4)} = x \cos x + 4 \sin x.$$

Question 2

First three non-zero terms of the Maclaurin series for $f(x) = \frac{2}{(2x^2 + 1)^{3/2}} = 2(1 + 2x^2)^{-3/2}$.

Successive differentiation. From $f = 2(1 + 2x^2)^{-3/2}$, $f' = -12x(1 + 2x^2)^{-5/2}$, which satisfies $(1 + 2x^2)f' + 6xf = 0$. Differentiating this successively and putting $x = 0$ at each stage ($f(0) = 2$):

$$(1 + 2x^2)f'' + 10xf' + 6f = 0 \Rightarrow f''(0) = -12,$$

$$(1 + 2x^2)f''' + 14xf'' + 16f' = 0 \Rightarrow f'''(0) = 0,$$

$$(1 + 2x^2)f^{(4)} + 18xf''' + 30f'' = 0 \Rightarrow f^{(4)}(0) = 360.$$

Hence $f(x) = f(0) + \frac{f''(0)}{2!}x^2 + \frac{f^{(4)}(0)}{4!}x^4 + \dots$:

$$f(x) = 2 - 6x^2 + 15x^4 + \dots$$

Alternative — binomial series. With $u = 2x^2$, $(1 + u)^{-3/2} = 1 - \frac{3}{2}u + \frac{15}{8}u^2 + \dots$; substituting and multiplying by 2 gives $2 - 6x^2 + 15x^4 + \dots$ directly.

Question 3

Using five terms of $y = \ln(2x + 4)$, approximate $\ln 4.02$ (7 d.p.).

Repeated differentiation. With $y = \ln(2x + 4)$, $y' = \frac{1}{x + 2}$, $y'' = -\frac{1}{(x + 2)^2}$, $y''' = \frac{2}{(x + 2)^3}$, $y^{(4)} = -\frac{6}{(x + 2)^4}$, so

$$y(0) = \ln 4, \quad y'(0) = \frac{1}{2}, \quad y''(0) = -\frac{1}{4}, \quad y'''(0) = \frac{1}{4}, \quad y^{(4)}(0) = -\frac{3}{8}.$$

Building $y = \sum \frac{y^{(n)}(0)}{n!}x^n$ to five terms:

$$\ln(2x + 4) = \ln 4 + \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{24} - \frac{x^4}{64} + \dots$$

For $\ln 4.02$ we need $2x + 4 = 4.02 \Rightarrow x = 0.01$:

$$\ln 4.02 \approx 1.3862943611 + 0.005 - 0.0000125 + 0.0000000417 - 0.0000000002 = 1.3912819.$$

Alternative — standard log series. $\ln(2x + 4) = \ln 4 + \ln\left(1 + \frac{x}{2}\right)$; expanding $\ln\left(1 + \frac{x}{2}\right) = \frac{x}{2} - \frac{x^2}{8} + \frac{x^3}{24} - \frac{x^4}{64} + \dots$ gives the same five terms.

Check. Exact $\ln 4.02 = 1.39128190263\dots$; the five-term estimate is correct to all 7 d.p. ✓

Question 4

Maclaurin series (first four non-zero terms).

(a) $f(x) = \sec x$. Differentiating twice, $f' = \sec x \tan x$ and $f'' = 2 \sec^3 x - \sec x$, i.e. $y'' = 2y^3 - y$ for $y = \sec x$. Differentiating this relation successively and evaluating at $x = 0$ ($y(0) = 1$, $y'(0) = 0$):

$$\begin{aligned} y'' &= 2y^3 - y & \Rightarrow y''(0) &= 1, \\ y''' &= y'(6y^2 - 1) & \Rightarrow y'''(0) &= 0, \\ y^{(4)} &= y''(6y^2 - 1) + 12y(y')^2 & \Rightarrow y^{(4)}(0) &= 5, \\ y^{(5)} &= y'''(6y^2 - 1) + 36yy'y'' + 12(y')^3 & \Rightarrow y^{(5)}(0) &= 0, \\ y^{(6)} &= y^{(4)}(6y^2 - 1) + 72(y')^2y'' + 36y(y'')^2 + 48yy'y''' & \Rightarrow y^{(6)}(0) &= 61. \end{aligned}$$

Hence

$$\sec x = 1 + \frac{1}{2}x^2 + \frac{5}{24}x^4 + \frac{61}{720}x^6 + \dots$$

Alternative — compare coefficients. Writing $\sec x = a_0 + a_2x^2 + a_4x^4 + a_6x^6 + \dots$ and matching coefficients in $\cos x \cdot \sec x = 1$ gives $a_0 = 1$, $a_2 = \frac{1}{2}$, $a_4 = \frac{5}{24}$, $a_6 = \frac{61}{720}$.

(b) $f(x) = \tan^{-1} x$. Here $y' = \frac{1}{1+x^2}$, i.e. $(1+x^2)y' = 1$. Differentiating successively and putting $x = 0$ at each stage ($y(0) = 0$, $y'(0) = 1$):

$$\begin{aligned} (1+x^2)y'' + 2xy' &= 0 & \Rightarrow y''(0) &= 0, \\ (1+x^2)y''' + 4xy'' + 2y' &= 0 & \Rightarrow y'''(0) &= -2, \\ (1+x^2)y^{(4)} + 6xy''' + 6y'' &= 0 & \Rightarrow y^{(4)}(0) &= 0, \\ (1+x^2)y^{(5)} + 8xy^{(4)} + 12y''' &= 0 & \Rightarrow y^{(5)}(0) &= 24, \\ (1+x^2)y^{(6)} + 10xy^{(5)} + 20y^{(4)} &= 0 & \Rightarrow y^{(6)}(0) &= 0, \\ (1+x^2)y^{(7)} + 12xy^{(6)} + 30y^{(5)} &= 0 & \Rightarrow y^{(7)}(0) &= -720. \end{aligned}$$

Hence $\tan^{-1} x = x + \frac{-2}{3!}x^3 + \frac{24}{5!}x^5 + \frac{-720}{7!}x^7 + \dots$:

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$$

Alternative — integrate a known series. $\frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots$; integrating term-by-term ($\tan^{-1} 0 = 0$) gives the same series.

(c) $f(x) = \sinh x$. The derivatives cycle: $y' = \cosh x$, $y'' = \sinh x$, $y''' = \cosh x, \dots$, so at $x = 0$, $y(0) = 0$, $y'(0) = 1$, $y''(0) = 0$, $y'''(0) = 1, \dots$. Hence

$$\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots = x + \frac{x^3}{6} + \frac{x^5}{120} + \frac{x^7}{5040} + \dots$$

Alternative — odd part of e^x . $\sinh x = \frac{1}{2}(e^x - e^{-x})$ retains the odd powers of $e^x = \sum x^n/n!$, giving the same series.

Question 5

$$f(x) = \cot^{-1} x.$$

(a) Here $f'(x) = -\frac{1}{1+x^2}$, i.e. $(1+x^2)f' = -1$. Differentiating successively and putting $x = 0$ at each stage ($f(0) = \frac{\pi}{2}$, $f'(0) = -1$):

$$\begin{aligned}(1+x^2)f'' + 2xf' &= 0 &\Rightarrow f''(0) &= 0, \\(1+x^2)f''' + 4xf'' + 2f' &= 0 &\Rightarrow f'''(0) &= 2, \\(1+x^2)f^{(4)} + 6xf''' + 6f'' &= 0 &\Rightarrow f^{(4)}(0) &= 0, \\(1+x^2)f^{(5)} + 8xf^{(4)} + 12f''' &= 0 &\Rightarrow f^{(5)}(0) &= -24.\end{aligned}$$

This gives $f(x) = \frac{\pi}{2} - x + \frac{2}{3!}x^3 + \frac{-24}{5!}x^5 + \dots$:

$$\cot^{-1} x = \frac{\pi}{2} - x + \frac{x^3}{3} - \frac{x^5}{5} + \dots$$

Alternative — shift of $\tan^{-1}$. Since $\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$, negating the series of 4(b) and adding $\frac{\pi}{2}$ gives the same result.

(b) Series for $\frac{1}{1+x^2}$. Since $f'(x) = -\frac{1}{1+x^2}$, differentiate the series of (a) term-by-term (extending the pattern by $+\frac{x^7}{7}$):

$$f'(x) = -1 + x^2 - x^4 + x^6 - \dots \quad \Rightarrow \quad \frac{1}{1+x^2} = 1 - x^2 + x^4 - x^6 + \dots$$

Alternative — geometric series. $\frac{1}{1+x^2} = \frac{1}{1-(-x^2)} = 1 - x^2 + x^4 - x^6 + \dots$ for $|x| < 1$.

Question 6

Using standard results — first four non-zero terms.

(a) $f(x) = \cos 3x$, from $\cos u$ with $u = 3x$:

$$\cos 3x = 1 - \frac{9}{2}x^2 + \frac{27}{8}x^4 - \frac{81}{80}x^6 + \dots$$

(b) $f(x) = \frac{x^2}{1-3x}$, from $\frac{1}{1-3x} = 1 + 3x + 9x^2 + 27x^3 + \dots$ ($|x| < \frac{1}{3}$):

$$\frac{x^2}{1-3x} = x^2 + 3x^3 + 9x^4 + 27x^5 + \dots$$

(c) $f(x) = \cosh\left(\frac{3x}{2}\right)$, from $\cosh u$ with $u = \frac{3x}{2}$:

$$\cosh \frac{3x}{2} = 1 + \frac{9}{8}x^2 + \frac{27}{128}x^4 + \frac{81}{5120}x^6 + \dots$$

Question 7

Four-term Maclaurin series for $f(x) = \sinh^{-1} x$; approximate $\sinh^{-1} 0.2$ (8 d.p.).

Here $y' = (1+x^2)^{-1/2}$, so $(1+x^2)(y')^2 = 1$; differentiating (and dividing by $2y'$) gives $(1+x^2)y'' + xy' = 0$. Differentiating this successively and putting $x = 0$ at each stage ($y(0) = 0$, $y'(0) = 1$):

$$\begin{aligned}(1+x^2)y'' + xy' &= 0 &\Rightarrow y''(0) &= 0, \\(1+x^2)y''' + 3xy'' + y' &= 0 &\Rightarrow y'''(0) &= -1, \\(1+x^2)y^{(4)} + 5xy''' + 4y'' &= 0 &\Rightarrow y^{(4)}(0) &= 0, \\(1+x^2)y^{(5)} + 7xy^{(4)} + 9y''' &= 0 &\Rightarrow y^{(5)}(0) &= 9, \\(1+x^2)y^{(6)} + 9xy^{(5)} + 16y^{(4)} &= 0 &\Rightarrow y^{(6)}(0) &= 0, \\(1+x^2)y^{(7)} + 11xy^{(6)} + 25y^{(5)} &= 0 &\Rightarrow y^{(7)}(0) &= -225.\end{aligned}$$

So $\sinh^{-1} x = x + \frac{-1}{3!}x^3 + \frac{9}{5!}x^5 + \frac{-225}{7!}x^7 + \dots$:

$$\sinh^{-1} x = x - \frac{x^3}{6} + \frac{3x^5}{40} - \frac{5x^7}{112} + \dots$$

At $x = 0.2$:

$$\sinh^{-1} 0.2 \approx 0.2 - 0.00133333 + 0.000024 - 0.00000057 = \boxed{0.19869010}.$$

Alternative — binomial then integrate. $(1+x^2)^{-1/2} = 1 - \frac{1}{2}x^2 + \frac{3}{8}x^4 - \frac{5}{16}x^6 + \dots$; integrating term-by-term ($\sinh^{-1} 0 = 0$) gives the same series.

Check. Exact $\sinh^{-1} 0.2 = \ln(0.2 + \sqrt{1.04}) = 0.19869011\dots$; the four-term value agrees to 7 d.p. ✓

Question 8

$y = \tanh x$.

(a) Show $y'' = -2y + 2y^3$. Since $\frac{d}{dx} \tanh x = \operatorname{sech}^2 x = 1 - \tanh^2 x$:

$$y' = 1 - y^2 \Rightarrow y'' = -2y y' = -2y(1 - y^2) = \boxed{-2y + 2y^3}.$$

(b) Differentiate up to $y^{(5)}$ and build the series. Evaluating at $x = 0$ (where $y = 0$, $y' = 1$):

$$\begin{aligned} y' = 1 - y^2 &\Rightarrow y'(0) = 1, & y'' = -2y + 2y^3 &\Rightarrow y''(0) = 0, \\ y''' = y'(-2 + 6y^2) &\Rightarrow y'''(0) = -2, & y^{(4)} = y''(-2 + 6y^2) + 12y(y')^2 &\Rightarrow y^{(4)}(0) = 0, \end{aligned}$$

$$y^{(5)} = y'''(-2 + 6y^2) + 36yy'y'' + 12(y')^3 \Rightarrow y^{(5)}(0) = 4 + 12 = 16.$$

Substituting into $y = \sum \frac{y^{(n)}(0)}{n!} x^n$:

$$\tanh x = x + \frac{-2}{3!}x^3 + \frac{16}{5!}x^5 + \cdots = \boxed{x - \frac{x^3}{3} + \frac{2x^5}{15} + \cdots}.$$

Alternative — divide the series. Since $\tanh x = \frac{\sinh x}{\cosh x}$ and $\operatorname{sech} x = \frac{1}{\cosh x} = 1 - \frac{x^2}{2} + \frac{5x^4}{24} - \cdots$,

$$\tanh x = \left(x + \frac{x^3}{6} + \frac{x^5}{120} + \cdots\right) \left(1 - \frac{x^2}{2} + \frac{5x^4}{24} - \cdots\right) = x - \frac{x^3}{3} + \frac{2x^5}{15} + \cdots,$$

the same series without repeated differentiation.

Check. This is the standard expansion of $\tanh x$; the leading term x and alternating-sign cubic match $\tanh x \approx x$ for small x . ✓